

Date: May 13 2026 12:49AM
To: "Khair Razlan Othman" khair.othman@mmu.edu.my
From: "PLOS ONE" plosone@plos.org
Subject: PLOS ONE Decision: Revision required [PONE-D-26-06954]

PONE-D-26-06954

Evaluating the Effectiveness of Open-Source LLMs for Multi-Domain Bengali Text Classification with Zero-Shot and Few-Shot Prompting
PLOS One

Dear Dr. Othman,

Thank you for submitting your manuscript to PLOS ONE. After careful consideration, we feel that it has merit but does not fully meet PLOS ONE's publication criteria as it currently stands. Therefore, we invite you to submit a revised version of the manuscript that addresses the points raised during the review process.

Please submit your revised manuscript by Jun 27 2026 11:59PM. If you will need more time than this to complete your revisions, please reply to this message or contact the journal office at plosone@plos.org. When you're ready to submit your revision, log on to <https://www.editorialmanager.com/pone/> and select the 'Submissions Needing Revision' folder to locate your manuscript file.

Please include the following items when submitting your revised manuscript:

- A letter that responds to each point raised by the academic editor and reviewer(s). You should upload this letter as a separate file labeled 'Response to Reviewers'.
- A marked-up copy of your manuscript that highlights changes made to the original version. You should upload this as a separate file labeled 'Revised Manuscript with Track Changes'.
- An unmarked version of your revised paper without tracked changes. You should upload this as a separate file labeled 'Manuscript'.

If you would like to make changes to your financial disclosure, please include your updated statement in your cover letter. Guidelines for resubmitting your figure files are available below the reviewer comments at the end of this letter.

If applicable, we recommend that you deposit your laboratory protocols in protocols.io to enhance the reproducibility of your results. Protocols.io assigns your protocol its own identifier (DOI) so that it can be cited independently in the future. For instructions see: <https://journals.plos.org/plosone/s/submission-guidelines#loc-laboratory-protocols>. Additionally, PLOS ONE offers an option for publishing peer-reviewed Lab Protocol articles, which describe protocols hosted on protocols.io. Read more information on sharing protocols at https://plos.org/protocols?utm_medium=editorial-email&utm_source=authorletters&utm_campaign=protocols.

As the corresponding author, your ORCID iD is verified in the submission system and will appear in the published article. PLOS supports the use of ORCID, and we encourage all coauthors to register for an ORCID iD and use it as well. Please encourage your coauthors to verify their ORCID iD within the submission system before final acceptance, as unverified ORCID iDs will not appear in the published article. *Only* the individual author can complete the verification step; PLOS staff *cannot* verify ORCID iDs on behalf of authors.

We look forward to receiving your revised manuscript.

Kind regards,

Anirban Bhowmick, Ph.D.
Academic Editor
PLOS One

Journal Requirements:

When submitting your revision, we need you to address these additional requirements.

1. Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming. The PLOS ONE style templates can be found at

https://journals.plos.org/plosone/s/file?id=wjVg/PLOSOne_formatting_sample_main_body.pdf and

https://journals.plos.org/plosone/s/file?id=ba62/PLOSOne_formatting_sample_title_authors_affiliations.pdf

2. Please note that PLOS One has specific guidelines on code sharing for submissions in which author-generated code underpins the findings in the manuscript. In these cases, we expect all author-generated code to be made available without restrictions upon publication of the work. Please review our guidelines at <https://journals.plos.org/plosone/s/materials-and-software-sharing#loc-sharing-code> and ensure that your code is shared in a way that follows best practice and facilitates reproducibility and reuse.

3. We note that the grant information you provided in the 'Funding Information' and 'Financial Disclosure' sections do not match.

When you resubmit, please ensure that you provide the correct grant numbers for the awards you received for your study in the 'Funding Information' section.

4. Please ensure that you refer to Figure 2, 3, 4, 5, 6, 7, 8 and 9 in your text as, if accepted, production will need this reference to link the reader to the figure.

5. We note you have included a table to which you do not refer in the text of your manuscript. Please ensure that you refer to Table 4 in your text; if accepted, production will need this reference to link the reader to the Table.

6. If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited. There is no requirement to cite these works unless the editor has indicated otherwise.

[Note: HTML markup is below. Please do not edit.]

Reviewer's Responses to Questions

Comments to the Author

1. Is the manuscript technically sound, and do the data support the conclusions?

The manuscript must describe a technically sound piece of scientific research with data that supports the conclusions. Experiments must have been conducted rigorously, with appropriate controls, replication, and sample sizes. The conclusions must be drawn appropriately based on the data presented.

Reviewer #1: Yes

Reviewer #2: Partly

2. Has the statistical analysis been performed appropriately and rigorously?

Reviewer #1: Yes

Reviewer #2: No

3. Have the authors made all data underlying the findings in their manuscript fully available?

The [PLOS Data policy](#) requires authors to make all data underlying the findings described in their manuscript fully available without restriction, with rare exception (please refer to the Data Availability Statement in the manuscript PDF file). The data should be provided as part of the manuscript or its supporting information, or deposited to a public repository. For example, in addition to summary statistics, the data points behind means, medians and variance measures should be available. If there are restrictions on publicly sharing data—e.g. participant privacy or use of data from a third party—those must be specified.

Reviewer #1: Yes

Reviewer #2: No

4. Is the manuscript presented in an intelligible fashion and written in standard English?

PLOS ONE does not copyedit accepted manuscripts, so the language in submitted articles must be clear, correct, and unambiguous. Any typographical or grammatical errors should be corrected at revision, so please note any specific errors here.

Reviewer #1: Yes

Reviewer #2: Yes

5. Review Comments to the Author

Please use the space provided to explain your answers to the questions above. You may also include additional comments for the author, including concerns about dual publication, research ethics, or publication ethics. (Please upload your review as an attachment if it exceeds 20,000 characters)

Reviewer #1: 1. There is a serious inconsistency in the experimental hardware configuration. The previous description stated that a single RTX 4060 8GB graphics card was used to run the 72B model. However, even with 4-bit quantization, the 72B model still requires approximately 36GB of VRAM. 8GB of VRAM plus CPU offloading is insufficient to complete the full inference within a reasonable timeframe. Please re-verify the experimental environment and clarify whether cloud computing power or a multi-GPU cluster was actually used.

2. The cue word template design for the zero-shot and few-shot experiments lacks methodological support. The paper does not specify whether the cue words were chosen in Bengali or English, nor does it report any pre-experimentation or calibration of the cue words. This directly affects the fair comparison baseline for all models.

3. Five samples are used consistently in the few-shot experiments, but the selection strategy for these five samples is not explained. Are they randomly selected from the training set or selected based on specific criteria? Different choices may lead to performance fluctuations of over 10%. Please supplement the specific method for sample selection and the random seed.

4. The sentiment analysis task suffers from a severe class imbalance. The SentiNoB dataset contains 654 positive samples, 571 negative samples, and only 361 neutral samples. The model's F1 score for the neutral class is generally

low, which the paper attributes to insufficient model capability, but fails to control for the confounding variable of data imbalance.

5. In the fake news detection task, the training data was downsampled, but the specific sample size after downsampling was not specified, nor was it verified whether downsampling changed the semantic distribution characteristics of the original data. This could lead to an overestimation of model performance.

6. All models use the exact same temperature parameter 0 and top-p 0.8. This is unfair to the instruction-tuned model and the base model, as the base model is more prone to format errors or refusal to answer in deterministic output mode. Please design differentiated decoding strategies for different model types.

7. The study did not report any statistical significance tests. Differences in all performance metrics rely solely on numerical comparisons, making it impossible to determine whether the difference between Qwen and Gemma is statistically significant. Please supplement the results of paired tests or resampling tests.

8. The test set sizes for the four classification tasks vary significantly, ranging from 1586 samples for sentiment analysis to 1000 samples for hate speech detection and 625 samples for emotion recognition. Small test sets result in overly wide confidence intervals for the F1 score. Please report the 95% confidence interval for each task.

9. The error analysis section only shows the confusion matrices of the best and worst models, without providing linguistic analysis of common error patterns. For example, it does not analyze whether the model produces systematic misclassifications due to Bengali kinship terms or honorific systems.

10. The study lacks any baseline comparisons. It does not compare with traditional Bengali pre-trained models such as BanglaBERT, nor with API-based commercial models such as GPT-4. This makes it impossible for readers to determine the true relative performance of the open-source LLM. Please provide at least one strong baseline model.

You can refer to and learn from these references (Not published by myself or a collaborator)

Bu, X. (2025). MAPFusion: Enhancing RoBERTa for news classification with multi-head attention pooling and feature fusion. *International Scientific Technical and Economic Research*, 3(4), 71-84. <https://doi.org/10.71451/ISTAER2555>

Sheng, D., Ma, Z., Feng, H., Zhou, C., Liu, H., Guo, H., & Sheng, C. (2025). Comment analysis of online shopping based on big data and NLP techniques. *International Scientific Technical and Economic Research*, 3(2), 1-6.

<https://doi.org/10.71451/ISTAER2515>

Zhong, M., & Zhong, X. (2025). Online music personalized recommendation algorithm based on KKBox large-scale real-world dataset and integrated behavior sequence transformer. *International Scientific Technical and Economic Research*, 3(3), 116-134. <https://doi.org/10.71451/ISTAER2546>

Chen, G. (2026). Research on algorithm improvement of ARIMA-LSTM hybrid model in time series prediction of inflation rate. *International Scientific Technical and Economic Research*, 4(1), 90-122. <https://doi.org/10.71451/ISTAER2605>

Wang, W., Shen, S., & Wang, Y. (2026). Forecasting short-term export volumes with hybrid models integrating SARIMA with attention-based LSTM. *International Scientific Technical and Economic Research*, 4(1), 1-22.

<https://doi.org/10.71451/ISTAER2601>

Reviewer #2: The study described in the manuscript presents the results of an evaluation of six open-source large language models (LLMs), which differ primarily in terms of their size and pre-training datasets, as applied to four Bengali text classification tasks.

I have found a few issues in the manuscript that require further review before it can be published. I will list them in order of importance, from most to least:

1) Presentation and some inaccurate information (or information lacking supporting references)

- Section "Progress in Bengali LLMs and Benchmark Development" (line 130): Organize or synthesize its information in a table.

- Table 1: In the Test Size column, what are units have been chosen? Words, sentences, tokens, examples...

- Table 1: In the Fake News Detection task, the Test Size column shows "Balanced" instead of a number.

- Emotion Recognition (line 259): It says there are 625 samples, but it doesn't specify whether this is a balanced dataset.

- Table 2: The Architecture for DeepSeek-R1-8B shows "Transformer + RL", but RL is a training/alignment method, not an architectural component like decoder, encoder, attention, or MoE.

- Fake News Detection (line 266): In line 269, I read "Since it is quite imbalanced, downsampling was used on real news samples", but it does not explain how the downsampling has been performed or what ratio has been used.

- Mistral-V3-7B (lines 349-351): "The instruction tuning and reinforcement learning from human feedback (RLHF) are employed after the model undergoes pretraining." This is only for the instruct-fine-tuned model, a variant of Mistral-V3-7B (Mistral-7B-Instruct-v0.3) Decoder-only Transformer, instruction-tuned.

- DeepSeek-R1-8B (line 356):

This section It merges the description of DeepSeek-R1 and its distilled version (DeepSeek-R1-8B) and oversimplifies the description of DeepSeek-R1. The full model is DeepSeek-R1, while the 8B version is DeepSeek-R1-Distill-Llama-8B, to improve reasoning capability, especially math, code, and STEM reasoning (DeepSeek describes R1 as targeting reasoning tasks); not for multilingual proficiency as indicated in the manuscript.

Line 361 displays "which includes 32 attention heads, 30 layers, and 4096 hidden dimension." The config says 32 attention heads, 32 hidden layers, and 4096 hidden size. So 30 layers is incorrect.

- Phi-4-14B (line 386):

In equation (6), q_{fact} doesn't represent the concept mentioned on line 401: "worth, q_{fact} represents logical density".

- Gemma-2-27B (line 414):

Line 421 displays "with attention heads of 64", is that right? Maybe 32?

Line 422 displays "a 8192 hidden dimension", is that right? Maybe 4608?

Equation (22): Is the base value correct? For Gemma-2 is 500000.

Line 440: The Gemma 2 technical report does not present Gemma-2-27B as being trained with this specific weighted multilingual loss. Do you have some supporting reference? Gemma 2 has multilingual ability, but the official report says the training data was primarily English and not specifically optimized for state-of-the-art multilingual capability.

Lines 441-443, equations (25) and (26): You show generic formulas, not verified or unsupported for Gemma 2.

Lines 444-445: "T denotes the temperature attribute, which is usually set at 0.7" Sure? In equation (26), this value would not upweight low-resource languages.

- Qwen-2.5-72B (line 447):

Line 455: "and 8194 hidden dimensions", is that right? Maybe 8192?

Line 464: "It is estimated that 12% of the tokens are assigned to the Indic language," Overestimated?

-Evaluation Metrics (line 529): "using contemporary evaluation metrics: Accuracy, Precision, Recall, and F1-score." Instead of contemporary, it would be better to say "stablished" or "common".

Line 531: "The F1 score is important for an imbalanced dataset," Better option: Macro-averaged F1 (Recommended when both classes matter equally) -> Computes F1 independently for each class and then averages them.

Lines 534-535: "And the most vital metric, Accuracy, gives an overall rate of the model's ability to correctly identify the labels based on overall predictions." In a binary (2-class) classification problem with class imbalance, accuracy should generally be avoided as the primary metric, because it can be misleading. A model can achieve high accuracy by always predicting the majority class while failing on the minority class.

- Few-Shot Performance (line 579): Line 588 says "A significant gain is achieved" but this does not hold true for all use cases or all large language models (LLMs).

- Tables 5 and 6:

Multi-class classification: Sentiment Analysis and Emotion recognition. Sentiment Analysis: The categories contain 1586 samples, where positive was 654, negative was 571, and neutral was 361. Emotion recognition (balanced or imbalanced?, it is not indicated in the manuscript): If the dataset is balanced, accuracy is often acceptable as a headline metric, because each class contributes similarly. Still, I'd report Macro-F1 (or per-class F1) alongside it, because accuracy can hide class-specific failures even when classes are roughly balanced. If the dataset is imbalanced, accuracy becomes much less meaningful: predicting majority classes well can dominate the score while minority classes are poorly handled. Prefer metrics that treat classes more equally or focus on minority-class performance: Macro-F1 (and per-class recall/F1).

- LLMs Performance Across Sentiment Categories (line 604):

Line 610 says "consistently showing that separating neutral sentiment is still difficult in the absence of context examples". Is the Neutral class perhaps underrepresented?

Line 613: "neutral class observed across several models". What happens with DeepSeek-R1-8B?

- LLMs Performance Across Emotion Categories (line 620):

Lines 626-627 say "Qwen-2.5-72B improves performance by 15% in the Disgust class compared to zero-shot prompting, it results in a 5% decrease in the Anger class." May these classes overlap in some cases?

- LLMs Performance Across Hate and Not-Hate Categories (line 632):

Line 636: "(e.g., Qwen-2.5-72B," is Qwen-2.5-72B in this group?

Lines 640-642: "This disparity indicates that smaller-sized or less multilingual models do not generalise to socially sensitive language situations in Bengali very well." Can it be compared to other languages?

Lines 644-651: They contain many inconsistent explanations, confuse "zero-shot" results with "few-shot" results, etc.

- LLMs Performance Across Fake and Real News (653):

Line 664: "such as LLaMA-3.2-3B and DeepSeek-R1-8B obtain slight improvements" but LLaMa-3.2-3B doesn't show any improvement; on the contrary, it performs even worse.

Line 668: "shows a marginal decrease from 80.5% to 81.2% for the fake class and from 85.4% to 86.2% for the real class." Decrease?

- Fig 14. and Fig 15. Some cells become unreadable with this color scheme.

- Lines 716-718: "This reflects a cautious prediction bias, where the models avoid labelling a text as hateful unless explicit aggression is present." Is there any example to support this?

- Receiver Operating Characteristic (ROC-AUC) Analysis (line 722):

In line 725 it is said "the positive and negative classes". Which classes are considered positive or negative in multi-class classification? How was the area under the ROC curve (AUC) calculated?

Line 728: "the sentiment analysis task reveals moderate separability for most models, with AUC values ranging from 0.63 to 0.72." Remember that:

0.7 - 0.79: Acceptable discrimination.

0.6 - 0.69: Poor discrimination.

- Discussion (line 752):

Lines 761-762: "Qwen-2.5-72B and Gemma-2-27B are uniformly superior on all tasks, demonstrating the joint benefits of model scale and high-quality multilingual pretraining." The advantage should not be attributed equally to multilingual pretraining: Qwen-2.5-72B has a stronger claim to multilingual pretraining, while Gemma-2-27B was trained primarily on English data and was not specifically optimized for state-of-the-art multilingual performance.

- Fig 17. Both figures must use the same scale on the Y-axis.

- Line 788: "Fake News detection is the most tractable task, with F1-scores over 84% across different models." Is that the case for all models?

- Limitations (line 841):

Line 844: "Details of the limitation of the workIn this work, the" This sentence needs to be reworded.

2) References

- line 284: Transformer decoder architecture [12], is [12] a good reference? The models considered in this study follow the decoder-only Transformer paradigm introduced by Vaswani et al. (2017) and popularized at scale by GPT-style models such as GPT-3 Brown et al. (2020).

- line 865: "Joboji" I can't find any reference to Joboji.

- References should include links whenever possible.

3) Typos

- line 71: "across four Bengali test classification" -> across four Bengali **text** classification

- lines 111-119: these lines refer to Section II through VI, the manuscript does not contain numbered sections

- line 132: "In **TinyLLAM**", Is there an "A" missing?

- line 172: Delete dot after NLP

- line 275: "In this research experiment was conducted on six" -> "In this research*,* experiment was conducted on six"

- line 305: Regarding equation (4), the following text states: "Here, the queries are organized into h headers, and the set g refers to a set of keys and 305 values, where $g < h$ ". However, g does not appear in equation (4) (which is the same formula as in equation 2).

- line 369: "reinforcing" -> **reinforcement**

- line 372: "polity" -> **policy**

- line 844: "Details of the limitation of the workIn this work, the" This sentence needs to be reworded.

4) Data availability: The datasets have been published in a repository, but the code has not been published, contrary to what is stated in the README file.

6. PLOS authors have the option to publish the peer review history of their article ([what does this mean?](#)). If published, this will include your full peer review and any attached files.

If you choose "no", your identity will remain anonymous but your review may still be made public.

Do you want your identity to be public for this peer review? For information about this choice, including consent withdrawal, please see our [Privacy Policy](#).

Reviewer #1: No

Reviewer #2: No

[NOTE: If reviewer comments were submitted as an attachment file, they will be attached to this email and accessible via the submission site. Please log into your account, locate the manuscript record, and check for the action link "View Attachments". If this link does not appear, there are no attachment files.]

To ensure your figures meet our technical requirements, please review our figure guidelines: <https://journals.plos.org/plosone/s/figures>

You may also use PLOS's free figure tool, NAAS, to help you prepare publication quality figures: <https://journals.plos.org/plosone/s/figures#loc-tools-for-figure-preparation>.

NAAS will assess whether your figures meet our technical requirements by comparing each figure against our figure specifications.

In compliance with data protection regulations, you may request that we remove your personal registration details at any time. ([Remove my information/details](#)). Please contact the publication office if you have any questions.